SOP-PRS-002: Pressure Release and Stored Energy Procedure

1. Purpose

This procedure describes how to safely release pressure from compressed air systems and liquid systems before maintenance. It is used together with SOP-LOTO-001: lockout isolates the energy sources, pressure release removes the energy that remains stored in the system.

2. Hazards

- Compressed air at 7 bar can blow off a loosened plug or filter cover with enough force to cause serious injury.
- A jet of compressed air directed at the skin can inject air into the body. **Never** use compressed air to clean clothing or skin.
- Hot liquid under pressure can flash to steam when released.
- Oil mist from a compressor separator vessel is flammable.

3. General rules

- Never loosen a flange, plug, filter cover or fitting on a system under pressure.
- Always use the designated vent or drain valves, never a loosened joint.
- Read the pressure gauge before and after venting. A gauge that reads 0 bar may be faulty; confirm by opening a vent valve and checking that no air or liquid escapes.
- Vent to a safe place, away from people, and wear eye protection and hearing protection when venting compressed air.

4. Compressed air systems (C-Series compressors)

1 Stop the compressor with the stop button. Wait until the motor has stopped; the controller unloads the compressor for 30 seconds before stopping.

2 Isolate and lock the main switch according to SOP-LOTO-001.

3 Close and lock the outlet isolation valve between the compressor and the air network. The network and air receivers stay pressurised; do not vent them unless the work requires it.

4 The automatic blow-off valve vents the separator vessel after stopping. Wait at least **2 minutes**.

5 Check the separator vessel pressure gauge. It must read **0 bar**.

6 Open the manual vent valve on the separator vessel to confirm that no pressure remains. Keep it open during the work.

7 When opening the oil filler plug, open it only one turn first and wait for any remaining pressure to escape.

4.1 Air receivers

If the receiver must be opened or worked on, isolate it from the network and the compressor, then open the receiver drain valve slowly until the gauge reads 0 bar. Leave the drain valve open during the work.

5. Liquid systems (P-Series pumps and pipework)

1 Stop the pump and lock out the motor according to SOP-LOTO-001.

2 Close and lock the discharge valve first, then the suction valve.

3 Check the liquid temperature. If it is above 60 °C, wait until it has cooled below 50 °C, or follow the hot work instructions for hot liquids.

4 Open the vent plug on top of the pump casing slowly, then open the casing drain plug and drain the liquid into a suitable container.

5 Confirm the discharge pressure gauge reads 0 bar and that no liquid flows from the open vent.

6 Check that the closed valves are holding: if liquid keeps flowing from the drain, a valve is passing. Stop work and isolate further upstream.

6. Returning to service

1 Close all vent and drain valves that were opened.

2 Remove locks according to SOP-LOTO-001.

3 For pumps: open the suction valve, fill and vent the casing before starting.

4 For compressors: open the outlet isolation valve slowly, then start the compressor.

7. Records

Record the pressure release on the work permit with the gauge readings before and after venting.